

Technical Document #137: Retrieval Augmented Generation - Comprehensive Overview

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Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Re-ranking models score retrieved documents for relevance to the query. They improve the quality of context provided to the generator.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Semantic search finds documents based on meaning rather than exact keywords. It uses embedding similarity to match queries with relevant content.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Semantic search finds documents based on meaning rather than exact keywords.
- Vector databases store dense embeddings for semantic search.
- Re-ranking models score retrieved documents for relevance to the query.